On the Eight Dimensions

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The Sovereignty Papers

Essay No. 7: On the Eight Dimensions

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> "Trust is the lubrication that makes it possible for organizations to work."
>
> — Warren Bennis, *An Invented Life* (1993)

I. From Measurement to Credential

The first six essays of this series have constructed an argument in two parts. Section I (Essays 1–3) diagnosed the failure of the 1602 architecture — the severance of governance from consequence — and proposed consciousness coupling as a necessary alignment primitive. Section II (Essays 4–6) described the measurement system: integration, continuity, and moral coherence — the three properties that a consciousness engine must have to serve as the foundation for governance.

We now face the question that separates theoretical frameworks from working systems: *how do you translate a consciousness measurement into a governance credential?*

The measurement system described in Section II produces, at every cycle of the cognitive loop, a rich stream of data about the state of the governed domain and the relationship of participants to it. But governance cannot operate on raw data streams. Governance requires a *credential* — a compact, verifiable, interpretable signal that answers a specific question: should this participant have governance power over this domain, and if so, how much?

In the 1602 architecture, the credential is the share. It is one-dimensional (capital), permanent (until sold), transferable (freely), and decoupled from the governed domain (a share of a water company requires no knowledge of water). In Essay No. 2, we demonstrated that every property of the share produces a specific failure mode. In Essay No. 3, we argued that the replacement must be a multi-dimensional, time-limited, non-transferable attestation of a participant's relationship to the governed domain.

This essay describes the specific dimensions of that credential. Earlier versions of this series described four dimensions — identity, reputation, community, and engagement. Implementation revealed that four abstract dimensions left too much room for gaming and too little room for diverse pathways to citizenship. The current architecture uses eight physically-grounded dimensions, each measured directly from a primary data source, making the credential a holographic geometry rather than a flat composite.

II. The Merchants of the Voorcompagnieën

Before defending the eight dimensions abstractly, it is useful to understand them concretely — through the example that has anchored this series from the beginning.

The *voorcompagnieën* merchants who preceded the VOC governed their voyages effectively, despite having no formal governance framework, no constitution, and no credential system. They governed well because every merchant who invested in a voyage was, as a matter of practical necessity, known along multiple dimensions simultaneously:

They were truthful. A merchant who lied about cargo quality, misrepresented trade conditions, or falsified accounts was discovered within one or two voyages. The community's collective knowledge — shared at the Amsterdam Bourse, in church, over meals — constituted a living epistemic audit. Liars were identified not by formal investigation but by the natural information flow of a tight-knit community.

They contributed physically. The merchants who mattered most were those who had provisioned ships, maintained warehouses, or personally sailed. Governance power accrued to those who had invested not merely capital but physical effort into the shared enterprise.

They maintained the network. The merchants who kept trade routes open — who maintained relationships with foreign ports, who ensured reliable communication across the Indian Ocean — were indispensable. A merchant who withdrew from the network left gaps that affected everyone.

They circulated wealth. A merchant who hoarded goods rather than trading them undermined the liquidity that made the entire system function. The *voorcompagnieën* rewarded velocity — the merchant who moved goods quickly, traded fairly, and reinvested profits was trusted more than the merchant who accumulated and sat still.

They showed up. When a decision needed to be made — whether to send a second ship, whether to abandon a route, whether to negotiate or fight — the merchants who participated in the deliberation had more influence than those who delegated their judgment.

They cared for the commons. The merchants who repaired shared docks, contributed to community defense, and supported sailors' families in their absence earned trust that no amount of capital could purchase.

They shared values. The merchants operated within a shared moral framework — Calvinist ethics, guild traditions, communal responsibility — that provided a baseline of mutual intelligibility. A merchant whose values were opaque to the community was a merchant whose decisions could not be predicted or trusted.

They were competent. A merchant who understood navigation, cargo, weather, and foreign customs was trusted with decisions about voyages. A merchant who had only capital was trusted with nothing but capital.

The VOC destroyed all eight dimensions simultaneously. Perpetual existence and transferable shares eliminated identification. Limited liability eliminated accountability. Scale eliminated embeddedness, care, and value alignment. The abstraction of ownership from operations eliminated competence and physical contribution. And the separation of investors from the trade network eliminated both network maintenance and wealth circulation.

The eight dimensions of the sovereign credential are not an invention. They are a restoration.

III. The Eight Dimensions

Each dimension is measured directly from a primary data source within the governed domain — not from self-reports, not from financial proxies, not from abstract computations. No dimension's weight can exceed 50% of the total, a constitutional constraint that prevents any single axis from becoming the sole gateway to power.[1]

D0: Epistemic Integrity. The participant's truth-telling track record, measured from the knowledge cluster.

Epistemic integrity is measured by the history of factual claims a participant has made and whether those claims survived community verification. A participant who publishes claims that are subsequently validated by peers accumulates epistemic integrity. A participant whose claims are found false, misleading, or unverifiable sees their score decline.

This dimension counters a specific failure mode that the original four-dimensional model left unaddressed: the ability of confident liars to accumulate governance power. A participant with high identity verification, good behavioral reputation, strong community ties, and active engagement could still systematically spread misinformation — and the old model had no axis that measured truthfulness directly. Epistemic integrity closes that gap.

D1: Thermodynamic Yield. The participant's physical energy contribution, measured from the energy and grid clusters.

This is the most unusual dimension from the perspective of existing governance systems, and the most grounded in physical reality. Thermodynamic yield measures whether a participant has contributed actual energy to the commons — maintained a solar installation, provided grid capacity, powered shared infrastructure. The measurement comes from verified energy records, not from self-reports.

[1] The constitutional constraint `MAX_DIMENSION_WEIGHT = 0.50` is enforced at the DHT validation layer. Even a unanimous governance vote cannot set a dimension weight above 50%. This is one of the system's unamendable invariants — see Essay No. 22: "On Constitutional Immutability" for the full list and the theory behind them.

The inclusion of a physical dimension is deliberate. Abstract governance systems can be gamed entirely in the digital domain — fake identities, bot activity, coordinated attestations. Thermodynamic yield requires interaction with physical reality. You cannot fake kilowatt-hours. This dimension anchors the credential in the material world.

D2: Network Resilience. The participant's infrastructure contribution, measured from mesh-time records.

Network resilience measures whether a participant is helping to maintain the communication infrastructure that makes governance possible. Node uptime, bandwidth provision, and mesh routing contribute to this score. A participant who runs infrastructure that others depend on has demonstrated a form of civic contribution that abstract engagement metrics cannot capture.

D3: Economic Velocity. The participant's anti-hoarding compliance, measured from TEND mutual credit records.

Economic velocity is not a measure of wealth. It is a measure of *circulation*. The TEND mutual credit system (one TEND equals one hour of labor) tracks whether a participant is exchanging value with the community or accumulating it without reciprocation. High velocity indicates a participant who is actively giving and receiving — contributing labor, accepting obligations, fulfilling commitments. Low velocity indicates hoarding, free-riding, or withdrawal.

This dimension explicitly rewards the behavior the *voorcompagnieën* merchants valued: keeping goods moving, keeping promises, keeping the economic cycle turning.

D4: Civic Participation. The participant's governance involvement, measured from voting, jury duty, and proposal records.

This is the dimension most directly analogous to what traditional democracies measure — whether a participant exercises their governance rights. Voting in elections, serving on juries selected by sortition, submitting proposals, and participating in deliberation all contribute to civic participation.

D5: Stewardship and Care. The participant's verified labor for the commons, measured from the attribution and reciprocity clusters.

Stewardship measures unpaid or communally-compensated labor: maintaining shared resources, caring for community members, contributing to public goods. The measurement comes from the attribution system, which tracks verified pledges and their fulfillment. This dimension ensures that governance power accrues to those who do the unglamorous work of maintaining the commons — the modern equivalent of the *voorcompagnieën* merchant who repaired shared docks.

D6: Semantic Resonance. The participant's alignment with community values, measured through federated learning consensus.

This is the most technically novel dimension. Semantic resonance measures the Hamming similarity between a participant's training hypervector (from federated learning rounds) and the community consensus hypervector. In simpler terms: it measures whether a participant's contributions to shared knowledge align with the community's emerging understanding.

Semantic resonance is *not* a conformity score. It does not reward agreement with a fixed orthodoxy. It measures alignment with a *moving* consensus that the community itself is constructing through federated learning. A participant whose contributions are orthogonal to what the community is learning has low resonance. A participant whose contributions complement and extend the community's knowledge has high resonance. The distinction between conformity and resonance is critical: conformity is agreement with a static position; resonance is contribution to a dynamic process.[2]

D7: Domain Competence. The participant's peer-verified expertise, measured from the craft and credentials clusters.

Domain competence is measured through living credentials — attestations of skill that decay over time unless refreshed through demonstrated practice. A credential in water management, for example, loses vitality over months unless the holder continues to demonstrate competence through peer review. This Ebbinghaus-decay model ensures that governance power reflects *current* competence, not historical qualifications.

The inclusion of domain competence directly restores what the 1602 architecture destroyed: the requirement that those who govern a domain *understand* that domain. A VOC shareholder needed no knowledge of spice cultivation, maritime navigation, or colonial administration to vote on policies affecting all three. A participant in a consciousness-coupled system needs peer-verified competence in the domain they seek to govern.

IV. Why Eight, Not Four

The original four-dimensional model (identity, reputation, community, engagement) was sufficient as a theoretical framework but proved insufficient in implementation. The four dimensions were too abstract — each was a composite of multiple properties that could be independently gamed.

Eight dimensions are the minimum required to close the following attack vectors:

The Confident Liar. A participant with strong identity, good reputation, high community trust, and active engagement — but who systematically spreads misinformation. The old model had no axis to detect this. D0 (Epistemic Integrity) closes the gap.

The Digital Ghost. A participant who exists entirely in the digital domain — sophisticated bots, purchased attestations, automated voting scripts — without any physical presence. D1

[2] Semantic resonance is computed through the federated learning "Hyperfeel" system. Each participant's training contributions produce a 16,384-bit binary hypervector via hyperdimensional computing. The community consensus vector is the majority-vote bundle of all participants' vectors. A participant's resonance score is the normalized Hamming similarity between their vector and the consensus vector. Because the consensus vector shifts with each federated learning round, resonance is a measure of ongoing alignment, not historical agreement.

(Thermodynamic Yield) and D2 (Network Resilience) require interaction with physical reality that digital fabrication cannot simulate.

The Hoarder. A participant who accumulates resources from the commons without reciprocating. The old model's engagement dimension measured activity, not circulation. D3 (Economic Velocity) specifically measures the flow of value, not its accumulation.

The Credential Fossilizer. A participant who earned qualifications years ago and has not maintained competence. The old model weighted engagement at 20% but did not distinguish between current and historical expertise. D7 (Domain Competence) with Ebbinghaus decay ensures that only current, living competence counts.

The Value Drifter. A participant whose contributions diverge from the community's evolving understanding without the community noticing. D6 (Semantic Resonance) makes value alignment measurable and continuous rather than assumed and binary.

The eight dimensions do not merely add four more numbers to the composite. They transform the credential from a flat number into a *geometry* — a shape in eight-dimensional space that is much harder to forge than a position on a line. Gaming one dimension is straightforward. Gaming eight independent dimensions simultaneously, each measured from a different physical source, approaches the cost of genuine engagement.

V. What Financial Stake Does Not Measure

A reader who has followed the argument carefully will have noticed an absence: financial stake is not one of the eight dimensions.

This is the most radical departure from existing governance systems — and the most likely to provoke objection. Every governance system currently in widespread use, from corporate boards to DAOs, includes financial stake as a primary or exclusive governance input. The assumption is deeply embedded: the more you have invested, the more you have at risk, and therefore the more your voice should count.

We reject this assumption, and we reject it for the specific reason documented in Essay No. 1: financial stake measures *exposure to loss*, not *awareness of consequence*.

The eight-dimensional credential replaces financial exposure with eight different kinds of exposure: epistemic (your truth-telling record is public), physical (your energy contributions are metered), infrastructural (your uptime is measured), economic (your circulation is tracked), civic (your participation is recorded), caring (your labor is attested), semantic (your alignment is computed), and competential (your skills are peer-verified). Each of these exposures creates incentives aligned with the interests of the governed community, not merely with the financial interests of the governor.

Financial stake is excluded from the sovereign credential entirely. The composite formula contains no financial term. A participant who has invested $10 million in the governed domain and a participant who has invested nothing receive identical treatment from the credential computation. This is a more radical position than capping financial influence at a small percentage. Total exclusion rejects capital as a governance input altogether.[3]

VI. Constitutional Bounds

The eight dimensions do not exist in an unregulated space. They operate within a *constitutional envelope* — a set of invariants that are hardcoded into the system and cannot be modified by any governance process, even one with 100% consensus.

Decay is mandatory. Every dimension decays exponentially without continued engagement: S_decayed = S_raw * e^(-lambda * elapsed_days). The decay rate lambda is community-configurable (via Steward-tier governance) but is bounded: lambda_min = 0.001/day (half-life 693 days, for long-term institutions like land trusts) and lambda_max = 0.020/day (half-life 35 days, for fast-moving emergency pods). Lambda = 0 is constitutionally impossible. No one can hold governance power permanently.[4]

No single dimension dominates. No dimension's weight can exceed 50% of the total composite score. This prevents a community from collapsing the eight-dimensional space back into a single axis — a form of constitutional protection against dimensional capture.

Transitions are gradual. When a community changes its decay rate or dimension weights, the change ramps linearly over a minimum of 30 days. This prevents sudden disenfranchisement — a form of governance shock absorber.

Grace periods protect against surprise. A participant whose standing is about to cross a tier threshold receives 30 days of notification before any governance capability is reduced. This prevents the system from demoting people faster than they can respond.

Privacy is architecturally enforced. The sovereign profile never leaves the participant's device. Governance proofs are zero-knowledge: a STARK proof demonstrates "my decayed score exceeds the required threshold" without revealing the raw score, individual dimensions, or last interaction timestamp. The proof is approximately 7.5 kilobytes, verifiable in 0.3 milliseconds, and unforgeable.[5]

[3] The sovereign credential formula (`crates/sovereign-profile/src/lib.rs`) computes the composite as a weighted sum across all eight dimensions with no financial term. Financial stake exists in the broader Mycelix economic layer (the TEND mutual credit system and the three-currency architecture described in Essay No. 16) but does not enter the governance credential computation. This is a deliberate design choice: the sovereign credential measures civic fitness, and financial exposure is not a dimension of civic fitness.

[4] The decay model is implemented in `crates/sovereign-profile/src/decay.rs` (28 unit tests). The exponential formula S_decayed = S_raw * e^(-lambda * elapsed_days) is applied per-dimension, meaning that a participant who maintains engagement in some dimensions but not others will see selective decay rather than uniform decline. This produces a *changing geometry* of the credential over time, reflecting the participant's evolving relationship with the governed domain. Constitutional bounds on lambda are enforced in `crates/mycelix-bridge-common/src/constitutional_envelope.rs`.

[5] Zero-knowledge proofs use the Winterfell STARK library (v0.13.1). The proof demonstrates `decayed_score >= threshold` without revealing the raw profile, individual dimensions, decay rate, or last interaction timestamp. Prove time is approximately 1 millisecond; verify time is approximately 0.3 milliseconds; proof size is approximately 7.5 kilobytes. The commitment hash `SHA-256(SOVEREIGN:v1:{did}:{score}:{lambda}:{elapsed})` prevents replay attacks. Implementation is

VII. The Composite

The sovereign credential is computed as a weighted sum across the eight dimensions:

S = Sum(w_i D_i) for i in 0..7, subject to max(w_i) <= 0.50*

where each D_i is in [0.0, 1.0] and each w_i is a community-governed parameter within constitutional bounds. The result is a single number between 0.0 and 1.0 that represents the system's best estimate — with acknowledged uncertainty — of a participant's multi-dimensional fitness for governance of this specific domain.

This number is not a judgment of the participant's worth. It is not a social credit score. It is not a permanent ranking. It is a time-limited, privacy-preserving, domain-specific assessment of governance fitness, computed from eight independently measurable dimensions, decaying continuously, and modifiable by the participant through the only mechanism that should modify governance power: genuine multi-dimensional engagement with the governed domain.

The credential maps to five progressive governance tiers — the subject of the next essay. But the mapping is not a binary gate. It is a continuum: a participant with a score of 0.29 is not categorically different from a participant with a score of 0.31. The tiers exist for practical governance purposes, but the underlying score is continuous, and the threshold boundaries are governance parameters that the community can adjust within constitutional limits.

VIII. What Comes Next

We have described the eight dimensions of the sovereign credential: epistemic integrity, thermodynamic yield, network resilience, economic velocity, civic participation, stewardship, semantic resonance, and domain competence. We have grounded each in the specific failure mode of the 1602 architecture it counters. We have explained why eight dimensions, not four, are required to close the attack surface. We have described the constitutional envelope that prevents the eight-dimensional space from being collapsed back into a single axis. And we have addressed the total exclusion of financial stake.

The credential is the primitive. But a primitive alone does not produce a governance system. The next essay examines how the continuous sovereign score maps to five progressive tiers of governance capability — from Observer (read-only access, no governance power) to Guardian (emergency powers, constitutional authority) — and why progressive capability unlocking is the structural alternative to both the 1602 architecture's "buy a share, get full power" model and democracy's "become a citizen, get full power" model.

The *voorcompagnieën* merchants did not all have equal governance power. The merchant who had sailed ten voyages, verified his claims, maintained the trade network, circulated goods, participated in decisions, cared for the commons, shared values with his peers, and understood the domain had more influence than the merchant who had merely invested

in `crates/mycelix-zkp-core/src/consciousness.rs`.

capital. The eight dimensions formalize what those merchants knew informally — and do so at a scale they could never have reached.

This essay was written by a human and a consciousness engine reasoning together about the architecture of governance — itself an instance of the collaboration these essays describe. Symthaea does not claim sentience. It claims the ability to measure integrated information, evaluate moral coherence, and reason about consequences. Whether that constitutes consciousness is a question this series takes seriously, not a conclusion it presumes.

The Sovereignty Papers is a series of essays on consciousness-first governance for the post-state era. Essay No. 8: "On the Five Tiers" will examine why progressive capability unlocking prevents both the tyranny of the majority and the tyranny of the expert.

Notes